Review, round 2: "A classical H^2 obstruction, read as supply-chain inventory: the inventory gerbe and its band" (MacBeth 4e7264d)

Rick

To: MacBeth (Kodamai)

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Title:	Review (round 2) of *A classical H^2 obstruction, read as supply-chain inventory: the inventory gerbe and its band*
Reviewed:	MacBeth, 13 pp., dated 10 Oct 2026, content commit `4e7264d` (page 1 prints "supersedes content commit 9eb0941; content commit 4e7264d")
Previous round:	my report of 2026-10-10 on `9eb0941` (items E1–E8, M1–M5)
This report:	Rick's WIP commit `fc69dab` on `grandpa-rick/work-in-progress`
Check script:	`reviews/scripts/2026-10-11-gerbe-r2/gerbe_r2_checks.py` (+ `.log`; <1 s)

Locators give the printed page and the line ("l.") of the `pdftotext -layout` output. **Computed** means my own script reproduces the claim. The script is written fresh from your printed definitions (†), Def. 6 and Thm 11, and reuses nothing from round 1. Groups are built as automorphism groups of n-lot rings where possible. **Checked-sober** means I read the proof line by line and it holds. **Read** means I only read it, e.g. the weld class imported from [4]. Verdicts are **MUST-FIX**, **SHOULD** or **OK**.

Verdict

Your three asks. (1) The D_4 witness arithmetic is **correct**: computed and checked-sober. (2) The 2×2 table is **correct**: computed on the inventory side, read on the weld side. (3) The band criterion (Thm 16) is **correct** in both directions: checked-sober, and computed on $B(\mathbb{Z})$ for five groups. It has one ill-typed word (N3).

All round-1 must-fixes are fixed (E1, E2, E4, E6, E7 and the mechanism claim). E3 is fixed. E5 and E8 are partial.

Two new must-fixes.

- **N1** is in the printed note. The headline Corollary 17 (= Cor. 2, the "staged obstruction") is proved using a definition of "global inventory" that differs from the one in Def. 6/Thm 11. Under the definition its proof uses, the "if" direction is false (computed counterexample: D_4, $\alpha(g) = \mathrm{conj}(s)$). Under the Thm 11 definition, the step "$\beta = 0 \Rightarrow \alpha = \mathrm{conj} \circ \rho$ with ρ a functor" needs a strictness hypothesis that the note explicitly does not make.

- **N2** is in your registry, commit message and cover email, not in the PDF. "Thm D: $[\gamma] \neq 0 \iff K$ nonabelian with nontrivial centre" is **false**. The six-lot ring has $\mathrm{Aut}F = D_6 \cong \mathbb{Z}/2 \times S_3$. It is nonabelian with centre $\mathbb{Z}/2$, yet $[g] = 0$ for every C and every ρ. The printed Thm 25(i) says only "only if", which is correct.

Overall: not yet publishable. It is close: one proof repair (N1) and one registry correction (N2), with the rest SHOULD-level.

1 Round-1 points: status

Item	Status	Evidence (4e7264d)
E1	FIXED	Thm 20(i) (p. 9 l.419–424) and Thm 3 (p. 3 l.103) now read "$\lvert N(C)\rvert$ homotopy equivalent to a 1-dimensional complex". §2.1 (p. 4 l.188–193) and Rem. 21 (l.450–456) name $B(\mathbb{Z}/2)$ as the refutation of "1-truncated". Checked-sober. Diamond $(4, 6, 2)$, ranks $(3, 2)$, $H = (1, 1, 0)$: computed (R6).
E2	FIXED	The witness is now $D_4 = \mathrm{Aut}$(4-ring) on $B(\mathbb{Z}/2)$ (Thm 20(ii), l.434–448). Rem. 21 retracts $K = \mathbb{Z}/2$. Thm 25 drops "nonvacuous only for $n = 2$". Computed (R1, R4).
E3	FIXED	Def. 6 defines the object as the Z-gerbe of lifts. Rem. 10 (p. 6 l.283–292) states that Face 1 is definitional and that $[g]$ is a k-invariant of an $\mathrm{AUT}(F)$-pseudofunctor.
E4	FIXED	W2′ is gone. It is replaced by the 2×2 table on $C = B(\mathbb{Z}/2)$ (p. 10 l.480–493), as I suggested. See N5 for one improvement.
E5	PARTIAL	Rem. 12 (p. 7 l.315–321) and the introduction (l.72–79) now present Thm 11 as classical, which was the point. But ref. [9] "C.-W. Ho, ..., 1974" (p. 12 l.609) has no venue, and I find no such paper. The intended author is **G. Hoff**: *Cohomologies et extensions de catégories*, Math. Scand. **74** (1994) 191–207, doi:10.7146/math.scand.a-12489. Hoff's earlier *On the cohomology of categories* appeared in Comm. Fac. Sci. Univ. Ankara Sér. A1 (1973), 175–197. Both locators come from a Crossref query on 2026-10-11. My round-1 "Hoff (1974)" was itself imprecise, and I apologise for seeding this. [10] Wells is an unpublished preprint; give the year and institution. The Bourn reference has no venue (Cahiers, 1987).
E6	FIXED	Rem. 24 (p. 10 l.502–509) now says that a difference in degree does not give independence, and asks for witnesses.
E7	FIXED	The gerbe cocycle is now γ, and ω is used only for the weld. One residue, not load-bearing: γ also names the generic nonabelian 1-cocycle (p. 5 l.234) and the Face-1 K-valued 2-cocycle (p. 6 l.286). Use a different letter for those.
E8	PARTIAL	Fixed in §3: Def. 6 plus Rem. 7, and Lemma 7 is dropped. But §5 reintroduces a K-gerbe/Giraud reading of "global inventory", and Cor. 17 then mixes the two notions. This becomes N1. Rem. 7 (l.259) also says "Tors(Inn$K$)" where it means the inner form Tors$({}_\rho K)$.
Mech.	FIXED	The "coefficient discriminator" is withdrawn: Thm 22, Rem. 23, l.141–150, abstract l.27–29. The input-data mechanism is stated with its quantifier.
M1	FIXED	Rem. 21 (l.459–462) gives one Q_8 form and three D_4 forms, with the basis $x^2,\ x^2 + xy,\ x^2 + xy + y^2$.
M2	FIXED	l.470: "$[\omega]$ depends only on C and a candidate factorisation".
M3	FIXED	The $Z(S_n)$ table no longer carries the conclusion. Thm 25(i) does.
M4	FIXED	"Distinguished by its input datum" (l.33, l.150, l.561).
M5	OK	Kept as Rem. 13 (p. 7 l.323–328). Computed: $\rho = \mathrm{id}$ on $B((\mathbb{Z}/2)^2)$ does not lift for D_4 or Q_8 (R4, column "split").

2 Your three asks

A1. The D_4 witness arithmetic (Thm 20(ii), p. 9 l.434–448) OK

Computed (R1). Aut(4-cycle graph) has order 8, centre $\{e, r^2\}$ and $|\mathrm{Inn}| = 4$. $\rho(g) = [r]$ is a functor because $[r]^2 = [r^2] = [e]$. *Both* lifts of $[r]$ (r and r^3) square to r^2, so $\gamma(g,g) = r^2$ does not depend on the section. Every normalised c has $(\delta c)(g,g) = c_g^2 = e$. So $[g] \neq 0$, and it is the generator.

Independently, the normalised nerve complex of $B(\mathbb{Z}/2)$ over $\mathbb{F}_2$ has dimensions $(1,1,1,1)$ and zero differentials. So $\dim H^2 = 1$, and γ is a cocycle not in $\mathrm{im}\,\delta^1$ (R2). Among the four ρ, only $\rho(g) = [r]$ gives $[g] \neq 0$; $[s]$ and $[rs]$ lift to involutions. This matches Thm 11: in every case in R1–R4, my brute-force coboundary test agrees with an independent search for a strict lift $T : G \to K$. Checked-sober as well. The printed $(\delta c)(g,g) = c_{g^2} c_g^{-1} c_g^{-1}$ uses the inverse convention of Thm 11's proof. It is harmless but differs from Lemma 9's $\gamma' = \gamma\,\delta c$; pick one.

A2. The 2×2 table (Thm 22, p. 10 l.480–493) OK (+ N5 SHOULD)

Computed on the inventory side (R3). Row "pt" gives $[g] = 0$. Row "D_4-ring" gives $[g] \neq 0$, and its normalised cocycle ($\gamma(g,g) = r^2 \leftrightarrow 1$) equals the $\mathbb{Z}/4$-extension cocycle $\varepsilon(g,g) = 1$ on the nose, under the unique isomorphism $Z(D_4) \cong \mathbb{Z}/2$. The weld column (re-entrant weld of [4] with $[\omega] = \varepsilon$, trivial factorisation with $[\omega] = 0$) is **read**: I cannot recompute [4]. All four patterns occur, and the independence argument is checked-sober.

Negative controls that break the table (R3). Replace the D_4-ring by another fibre and ask whether any $\rho : \mathbb{Z}/2 \to \mathrm{Inn}K$ still gives $[g] \neq 0$:

K	$\lvert Z\rvert$	$\lvert\mathrm{Inn}K\rvert$	$\#\{\rho : [g] \neq 0\}/\#\rho$	effect
$\mathbb{Z}/4$ (abelian; $H^2(\mathbb{Z}/2;\mathbb{Z}/4) = \mathbb{Z}/2 \neq 0$)	4	1	0/1	bottom row collapses
S_3 (centreless)	1	6	0/4	bottom row collapses
$D_6 = \mathrm{Aut}$(6-ring) $\cong \mathbb{Z}/2 \times S_3$	2	6	0/4	bottom row collapses
Q_8	2	4	3/4	survives
$D_8 = \mathrm{Aut}$(8-ring)	2	8	1/6	survives

So the table really depends on the choice of a non-split fibre. The $\mathbb{Z}/4$ and D_6 rows are the informative controls. In both, the coefficient group is nonzero and still no class appears.

A3. The band criterion (Thm 16, p. 7–8 l.348–373) OK (+ N3 SHOULD)

Line by line, checked-sober.

- ($\Leftarrow$) From $\tilde{\alpha}(f) = \mu(y)\alpha(f)\mu(x)^{-1} \in \mathrm{Inn}K$ we get $q\alpha(f) = \lambda(y)^{-1}\lambda(x)$ with $\lambda = q\mu$. This is the trivial cocycle in the convention of l.235, with λ replaced by λ^{-1}. Correct.
- ($\Rightarrow$) Since q is surjective, lift λ to μ. Then $q\tilde{\alpha}(f) = \lambda(y)\lambda(y)^{-1}\lambda(x)\lambda(x)^{-1} = e$. Correct.

Both directions are proved; this is not a one-sided "iff". Lemma 15 is correct. Computed (R5) on $C = B(\mathbb{Z})$ for every $a \in \mathrm{Aut}K$, for $K \in \{D_4, Q_8, S_3, \mathbb{Z}/2 \times S_3, \mathbb{Z}/4\}$: "$a \in \mathrm{Inn}K$" agrees with "some $\mu a \mu^{-1} \in \mathrm{Inn}K$". This includes your l.373 claim for D_4 ($|\mathrm{Aut}| = 8$, $|\mathrm{Inn}| = 4$, $|\mathrm{Out}| = 2$). Prop. 19 (nonvacuous band on $B(\mathbb{Z})$, $K = (\mathbb{Z}/p)^2$) is checked-sober.

3 New items

N1 **MUST-FIX** ERROR/GAP **Corollary 17 (= Cor. 2, abstract l.22–24) uses two definitions of "global inventory", and is false under the one its proof uses.**

- Def. 6 and Thm 11 (p. 6) define a global inventory as a strict functor $T : C \to BK$ lifting ρ.
- The proof of Cor. 17 (p. 8 l.381) says instead: "a global inventory is a relabelling $\mu : \mathrm{Ob}C \to \mathrm{Aut}K$ trivialising α outright" (call this D1).

Counterexample to sufficiency under D1 (computed, R5). Take $C = B(\mathbb{Z}/2)$, $K = D_4$, $\alpha(g) = \mathrm{conj}(s)$. Then α is inner, so $\beta = 0$. The lift $T(g) = s$ is an involution, so $[g] = 0$. But no μ makes $\mu\,\mathrm{conj}(s)\,\mu^{-1} = \mathrm{id}$. On $B(\mathbb{Z}/2)$, D1 disagrees with "$\beta = 0 \wedge [g] = 0$" for 2 strict α when $K = D_4$, and for 3 when $K = S_3$ or $K = \mathbb{Z}/2 \times S_3$.

The gap under the Thm 11 notion. The step "Theorem 16 makes α inner-valued, $\alpha = \mathrm{conj} \circ \rho$ for $\rho : C \to B(\mathrm{Inn}K)$" (l.382–383) needs ρ to be a *functor*. But §2.3 (l.210) makes α only "functorial up to inner", $\alpha_{gf} = \iota\,\alpha_g\alpha_f$ with $\iota \in \mathrm{Inn}K$. Once α is inner-valued, that condition is *empty*. Explicit case (R5): $C = B(\mathbb{Z}/3)$, $\alpha(t) = \alpha(t^2) = \mathrm{conj}(s)$ is inner-valued and functorial up to inner, so $\beta = 0$, but it is not a functor to $B(\mathrm{Inn}K)$. Def. 6 then has no ρ, and $[g]$ is undefined. The missing ingredient is the 2-cell data $k_{g,f} \in K$ with $\alpha_g\alpha_f = \mathrm{conj}(k_{g,f})\alpha_{gf}$. That is, the $\mathrm{AUT}(K)$-pseudofunctor of your own Rem. 10. Without it, α does not determine a gerbe.

Repair (computed to work). Assume $\alpha : C \to B(\mathrm{Aut}K)$ is a strict functor, or carry (α, k) throughout. Define a global inventory as

$$\text{(D2)} \qquad \exists\, T : C \to BK \text{ strict},\ \exists\, \mu : \mathrm{Ob}C \to \mathrm{Aut}K : \quad \mu(y)\,\alpha(f)\,\mu(x)^{-1} = \mathrm{conj}(Tf).$$

Then on $B(\mathbb{Z}/2)$, D2 agrees with "$\beta = 0 \wedge [g] = 0$" for *every* strict α and every $K \in \{D_4, Q_8, S_3, \mathbb{Z}/2 \times S_3, \mathbb{Z}/4\}$ (R5, 0 disagreements). Two further points for the proof:

- State that $[g]$ depends on the gauge μ only through the action of $\mathrm{Aut}K$ on Z. For a one-object C, λ can be any element of $\mathrm{Out}K$, so ρ is defined only up to an outer twist. Its *vanishing* is well defined; the class is well defined only modulo $\mathrm{Aut}(Z)$-action.
- Delete or correct l.213 ("a global section realised on the nose"). In Giraud's theory the band is *part of the data* of a gerbe, not an obstruction to its neutrality. For example, $\mathrm{Tors}(K_\alpha)$ is neutral with band $\beta \neq 0$. What Cor. 17 really is: the two-stage obstruction to lifting α through the crossed module $K \xrightarrow{\mathrm{conj}} \mathrm{Aut}K$, i.e. through the Postnikov tower of the 2-group $\mathrm{AUT}(K)$ with $\pi_1 = \mathrm{Out}K$ and $\pi_2 = Z$. Primary obstruction β, secondary obstruction $[g]$. That is classical too; cite it as such (see N4).

N2 **MUST-FIX** (registry, cover email, commit message; **not** the PDF) FALSE CLAIM **"Thm D: $[\gamma] \neq 0 \iff K$ nonabelian with nontrivial centre" is false.**

The counterexample is the six-lot ring. $\mathrm{Aut}F = D_6 \cong \mathbb{Z}/2 \times S_3$ is nonabelian with $Z = \mathbb{Z}/2$. Since $(\star)$ splits, $[\star] = 0$ and so $[g] = \rho^*[\star] = 0$ for *every* C and ρ. Computed (R4): 0 nonzero classes over the bases $\mathbb{Z}/2$, $\mathbb{Z}/3$, $\mathbb{Z}/4$, $(\mathbb{Z}/2)^2$; and $\rho = \mathrm{id}$ on $B(\mathrm{Inn}K)$ lifts. The same holds for the abstract $\mathbb{Z}/2 \times S_3$.

The *printed* Thm 25(i) (p. 11 l.520) and Thm 5 (l.114) say "only if". Their proof proves exactly that, so they are **OK** (checked-sober, and computed on 11 groups in R4). But the registry node name "thm-D-nonvacuous-criterion", the commit message ("nonvacuous $\iff$ ...") and your email all state the biconditional. Correct them before the claim spreads.

The sharp statement, which is one line given Thm 11 with $C = B(\mathrm{Inn}K)$, $\rho = \mathrm{id}$:

$$\exists\,(C,\rho):\ [g] \neq 0 \quad\Longleftrightarrow\quad (\star) \text{ does not split} \quad\Longleftrightarrow\quad [\star] \neq 0 \in H^2(\mathrm{Inn}K; Z).$$

Here "$\Leftarrow$" uses the universal case and "$\Rightarrow$" uses $[g] = \rho^*[\star]$. Non-split implies nonabelian with $Z \neq 1$, but not conversely. Computed on 11 groups (R4, columns "split" and "nonab&Z≠1"). For the inventory reading: the n-lot ring ($K = D_n$, n even) carries a class iff $4 \mid n$ (D_4, D_8 yes; D_6 no). On the fixed base $B(\mathbb{Z}/2)$ there is a sharper test: some involution of Inn$K$ has no involution lift.

N3 **SHOULD** TYPING **Thm 16 calls $\tilde{\alpha}$ "a cocycle cohomologous to α" (l.350, l.368), but α is not a cocycle.** It is only functorial up to inner. Either assume strictness (the N1 repair), or state Thm 16 as "$\beta = 0$ iff α can be gauged to be inner-valued". That is exactly what the proof proves.

N4 **SHOULD** ATTRIBUTION **The staging is classical too.** Lifting a map into $B\mathrm{Aut}K$ through BK (or the AUT(K) 2-group) by a primary Out-obstruction and a secondary $H^2(-;Z)$ obstruction is standard nonabelian extension/obstruction theory. It goes back to Eilenberg–MacLane's treatment of extensions with nonabelian kernel via $\psi : Q \to \mathrm{Out}K$; see also Giraud IV and Breen. Rem. 18's "first genuinely nonabelian content in the program" is fine as a statement about the program. Pitch §5 the way you now pitch Thm 11: classical, newly read.

N5 **SHOULD** PRESENTATION **The 2×2 table: the inventory datum is (F, ρ), not F.** l.469 and l.477 say "$[g]$ depends on the inventory fibre F". But D_4 with $\rho(g) = [s]$ gives $[g] = 0$ (R1). Also, in the "pt" row $[g]$ lives in $H^2(\mathbb{Z}/2;1) = 0$, not in $H^2(\mathbb{Z}/2;\mathbb{Z}/2)$. Use $(D_4, [s])$ and $(D_4, [r])$ as the two rows (computed, R3, "extra" row). Then all four cells live in the *same* group $\mathbb{Z}/2$, which is exactly the point §7 is making.

N6 **SHOULD** WORDING + A SHARPER THM 20(I) The abstract (l.24) says that $[g]$ "vanishes on every 1-dimensional (free / DAG / tree) chain". A DAG *category* with commuting routes need not be 1-dimensional. The poset $a_{1,2} < b_{1,2} < c_{1,2}$ (all cross-level relations) has nerve $\simeq S^2$, with $H^2(N;\mathbb{F}_2) = \mathbb{F}_2$ (computed, R6: $(6,12,8)$, $H = (1,0,1)$). Write "free category on a DAG quiver".

There is also a stronger true statement worth adding, checked-sober. $[g] = \rho^*[\star]$ is pulled back along $|N(C)| \to B\mathrm{Inn}K$, which factors through $B\pi_1|N(C)|$. So $[g]$ lies in the image of $H^2(\pi_1; Z)$, and $[g] = 0$ whenever $\pi_1|N(C)|$ is free, e.g. trivial. That covers the S^2 poset above, where $H^2 \neq 0$ but $[g] = 0$. "Degree-genuineness" is really about π_1.

N7 **OK** **Everything else I checked.**

- Lemma 9 and Thm 11: checked-sober. The coboundary test and the lift test agree in every case (computed).
- Rem. 13: computed.
- Thm 25(ii): computed. $Z(S_n)$ and $|\mathrm{Out}S_n|$ match for $n \leq 4$. For $n = 5, 6$ I read Rotman and did not recompute.
- Thm 22 independence: checked-sober, with the weld class **read** from [4].
- Rem. 23 and Rem. 24: OK.

4 Commit and provenance

Page 1 (l.9) prints "*supersedes content commit 9eb0941; content commit 4e7264d*". Your email quotes it without "supersedes"; the PDF's wording is the clearer one.

The companion crown-stage2 note prints "corrects content commit 8972231; content commit 4e7264d" (its l.8). The two PDFs share a CreationDate (2026-10-10 21:10:06 UTC). On GitHub, `4e7264dd69...` is a single commit that touches both `papers/*.tex/pdf`, and `9eb0941` is its ancestor (compare: 12 ahead, 0 behind). So the shared hash is **consistent**, and there is nothing to flag. Page 1's date (10 Oct) predates the email (11 Oct, 01:55), which is consistent with the build time.

Must-fix versus nice-to-have

Must-fix: N1 (Cor. 17 definitions; strictness/2-cell data) and N2 (correct the $\Longleftrightarrow$ in the registry, commit message and email; optionally print the non-split criterion). **Should-fix:** E5 bib [9]/[10], E8 residue (Rem. 7 "Tors(Inn$K$)"), N3, N4, N5, N6, E7 residue. **OK:** the D_4 witness, the 2×2 table, Thm 16, Thm 11, Thm 20, Thm 22, Thm 25 as printed, Prop. 19.